

Shutong Liu



✉ 23373306@buaa.edu.cn • ✉ andaul951@gmail.com
🌐 shutongliu-dev.github.io • 📞 +86 15191466082

Research Interests: Embodied Intelligence, Humanoid Foundation Models

Education

Beihang University | School of Computer Science | Computer Science and Technology 2023.9 – 2027.7
GPA: 3.87 / 4.0 **Rank: 3 / 198 (Top 1.5%)**
National Scholarship, University Outstanding Student, University Scholarship for Academic Excellence, Outstanding Student Cadre.

Research Experience

Whole-Body Tactile-Enhanced Humanoid Robot VLA Research

Uni-VLaT: Whole-Body Tactile Adaptation of VLA Policies for Humanoid Loco-Manipulation

(Under Review at IEEE ICRA 2027)

First Author | Advisor: Mengdi Xu (Institute for Interdisciplinary Information Sciences, Tsinghua) 2026.07 – Present

- **Project Content:** To address visual occlusion and the difficulty of perceiving body-contact states in contact-intensive whole-body manipulation tasks for humanoid robots, this project studies the integration of distributed whole-body tactile sensing into a pretrained Humanoid VLA model, improving the robot’s understanding of complex contact states, action adaptability, and task generalization. Uni-VLaT achieves a **75%** average success rate on **five real-robot tasks** on Unitree G1, improving over baselines without tactile sensing and with tactile input but without predictive supervision by **43%** and **7%**, respectively.
- **Core Method:** Design a spatiotemporal tactile encoder that uses region-wise encoding, learnable-query-based cross-region attention aggregation, and a temporal Transformer to encode four consecutive frames of tactile signals from eight whole-body regions into tactile tokens. A learnable gate connects the tokens to the DiT action head of the pretrained VLA, where they interact with proprioceptive states, actions, and vision-language features. Construct a tactile-anchored JEPA-style multimodal predictive auxiliary objective: using the contextualized tactile representation output by DiT as shared context, predict tactile signals, proprioceptive states, and visual latent representations for the next four steps; use EMA tactile and proprioceptive encoders and a frozen visual encoder to provide target representations, and jointly optimize the prediction loss with the Flow Matching action loss, guiding the policy to learn the relationships among contact, body response, and scene changes.
- **Personal Contribution:** As the first author, responsible for the overall project design, real-robot data collection, model training, and real-robot validation; focused on tactile representation learning and built the complete technical pipeline from data collection and policy training to deployment on a real robot.

Safety Alignment Research on Agentic Deception

Mitigating Agentic Deception via Evaluation-Side Partially Observable Games

(Under Review at ICLR 2027)

Second Author | Advisor: Xianglong Liu (School of Computer Science, Beihang University) 2026.03 – 2026.07

- **Project Content:** As large-model capabilities improve, deceptive behavior by large models and agents increasingly emerges across scenarios in RLHF. This work attributes it to partial observability on the evaluation side and proposes evaluation-side partially observable extensive-form games (e-PoG), constructing a max–min framework: the Agent maximizes reward, the Monitor reveals deception through complete interaction trajectories and reduces the Agent’s pay-off, and a parameter-frozen Verifier verifies accusations and converts them into penalties. We analyze the deception-suppression mechanism and theoretical guarantees from a game-equilibrium perspective and design an Agent–Monitor two-timescale alternating RL method, combining NLI verification with evidence checking to improve supervision reliability. On Gemma3-4B-IT and Qwen3-4B/8B across role-playing, SearchQA, and WebShop tasks, the method outperforms

RL and supervised baselines, mitigates multiple emergent deceptive behaviors while maintaining task performance, and remains effective when a weak model supervises a strong model.

- **Personal Contribution:** As the second author, responsible for task-scenario design, training framework and pipeline construction, and large-scale RL training based on GRPO and related algorithms.

RBA Defense via Dynamic Browser Fingerprints

From Frozen Pose to Live Dance: Stochastic Browser Fingerprinting for Robust Risk-Based Authentication

(Under Review at USENIX Security 2027)

First Author | Advisor: Shujiang Wu (School of Computer Science, Beihang University)

2025.07 – 2025.11

- **Project Content:** To address static replay attacks in risk-based authentication (RBA), propose a dynamic browser fingerprinting method. Construct dynamic challenges with the Canvas, WebGL, and Audio APIs; implement rendering challenges through pixel and subpixel position changes; extract stable features from responses and construct fingerprint hashes for browser identification based on dynamic challenge responses. Conduct cross-browser experiments on 213 devices. Among valid samples, the dynamic stability rates of Canvas, WebGL, and Audio fingerprints are **99.0%**, **91.7%**, and **100%**, respectively, effectively addressing real-world replay attacks in RBA and providing a new direction for browser fingerprinting.
- **Personal Contribution:** As the first author, proposed the dynamic fingerprinting defense idea, completed experimental investigation and validation, paper writing, and subsequent engineering application. The resulting work is currently used by the Beihang University Network Center.

Multimodal LLM Long-Document Modeling Research

HIPPOGIST: Visual Text Compression and Memory for Long-Horizon LLM Agents

Core Member | Advisor: Richong Zhang (School of Computer Science, Beihang University)

2025.11 – 2026.03

- **Project Content:** The work on DeepSeek-OCR and Zhipu Glyph suggests converting long text into images and processing it with a VLM to achieve token compression. Based on this idea, we analyze Attention patterns and regularities in visual long-document modeling and propose an Agent-based long-document processing architecture through empirical analysis, exploring support for ultra-long contexts and token compression while retaining model understanding. The framework splits long documents into chunks and renders them as images; in each round, the VLM reads the current image together with the previous summary, calls visual tools such as highlighting and zooming as needed, updates the summary with newly acquired information, and passes it to the next round before answering questions from the accumulated summary. We design a reinforcement learning algorithm to optimize multi-round reading and summary-maintenance policies and pass reward signals, providing a new approach to multimodal long-context problem solving.
- **Personal Contribution:** As a core project member, built the visual agentic long-document processing approach and completed all code implementation, including Agent pipeline construction and DAPO-based RL training on Qwen3VL-8B. Implemented the dual-memory maintenance and tool-calling workflow, trained with the ver1-agent framework on four A800 GPUs, used final-answer rewards to optimize the policies for tokens generated in preceding rounds, and trained cross-round memory decisions.

Smart Education Diagnosis and Recommendation Agent Project

Core Member | Advisor: Richong Zhang (School of Computer Science, Beihang University)

2025.07 – 2025.11

- **Project Content:** For the iFlytek problem in the Challenge Cup “Bounty Challenge,” built an intelligent Agent system for education that supports learning-status diagnosis, adaptive learning-path recommendation, and dynamic knowledge-graph construction; used LlamaFactory fine-tuning based on results from the project group to suppress hallucinations.
- **Personal Contribution:** Built an Agent interaction system with the LangGraph framework, designed the knowledge graph and system evaluation mechanism, and completed front-end and back-end code. Served as the primary coder and architect of the project. The work won the track’s Special Prize and championship in the Challenge Cup.

Awards

National Challenge Cup Finals, Special Prize (1st Place)	2025
China International College Students' Innovation Competition (CICSIC), National Bronze Award	2025
Capital Challenge Cup, Special Prize and Second Prize	2026
“Wukong Cup” CCF Quantum Programming Challenge, Team Third Prize	2025
The Chinese Mathematics Competitions for College Students (CMC), First Prize	2023, 2024
Contemporary Undergraduate Mathematical Contest in Modeling (CUMCM), National Second Prize	2025
China High School Mathematics League, Provincial First Prize	2022
Mathematical Contest in Modeling (MCM), Honorable Mention	2026

Technical Experience

- **Embodied Intelligence and Robot Learning**
Experienced with the Unitree G1 humanoid robot platform, the SONIC whole-body control framework, and tactile sensing systems, including humanoid-robot teleoperation, whole-body tactile data collection, multimodal sensing fusion, and contact-interaction task design. Familiar with robot simulation environments such as MuJoCo and Isaac Lab, with experience in simulation-scene construction, simulation-data generation, robot-model configuration, and reinforcement-learning-driven motion tracking.
- **Agent Systems and LLM Applications**
Experienced in LLM Agent system development, including Agent Harness design, context and long-term memory management, retrieval-augmented generation (RAG), and multi-agent collaboration workflows. Research experience in agent reliability, including agent deception analysis and reinforcement-learning alignment (RLHF) methods.
- **Algorithms and Engineering Tools**
Proficient in Python, C++, and PyTorch; familiar with the theory of RL, Transformers, and diffusion models. Familiar with Linux, Git, Docker, ROS2, and related systems and development tools.

Profile

During high school and university, won provincial first prizes in mathematics competitions. Have a strong mathematical foundation and reasoning ability, with solid knowledge of linear algebra, combinatorics, and related fundamentals. Have a broad understanding of theoretical knowledge in machine learning and large models, as well as hands-on experience with reinforcement learning and Agent engineering.

Current research experience includes a Web security paper, a multimodal long-document modeling project, an AI Deception safety-alignment RLHF project, and a project on Humanoid VLA models enhanced with whole-body tactile sensing. Familiar with building Agentic AI frameworks and managing memory systems.

Research interests focus on **foundation models for humanoid robots**, including **Behavior Foundation Models, World Action Models, reinforcement-learning-driven Whole-body Control, and Human-Object Interaction**. Aim to combine large-scale behavioral data, world-action models, and reinforcement learning to explore humanoid robots with long-term generalization and autonomous adaptation.

Passionate about research, curious and exploratory, diligent and motivated, and committed to connecting fundamental theory with practice. Believe in long-term commitment.